

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

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COURSE NAME: Computer Networks

COURSE CODE: CSA0704

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

2) AIM:

To create two LANs connected through a router using Cisco Packet Tracer, configure IPv4 addressing and default gateways, verify communication using the ping command, and analyze IPv4 header fields using Wireshark.

SOFTWARE REQUIRED: Cisco Packet Tracer and Wireshark.

PROCEDURE:

1. Open **Cisco Packet Tracer** and create two LANs using two PCs, two switches, and one router.
2. Connect **PC0** → **Switch0** → **Router0** → **Switch1** → **PC1** using suitable cables.
3. Configure PC0 with IP 192.168.1.2, subnet mask 255.255.255.0, and gateway 192.168.1.1.
4. Configure PC1 with IP 192.168.2.2, subnet mask 255.255.255.0, and gateway 192.168.2.1.
5. Configure Router0 interfaces with 192.168.1.1/24 and 192.168.2.1/24, and enable them using no shutdown and Verify that all network links are active and green.
6. From PC0, use the command ping 192.168.2.2 to verify end-to-end connectivity.
7. Open **Wireshark**, select the active network interface, and start packet capture.
8. Generate ICMP traffic using a ping, stop the capture, and apply the icmp filter.
9. Select an ICMP packet and expand **Internet Protocol Version 4** to analyze the **Source IP, Destination IP, TTL, Protocol Number, and Header Length**.

IMPLEMENTATION:

SCREENSHOT: Topology

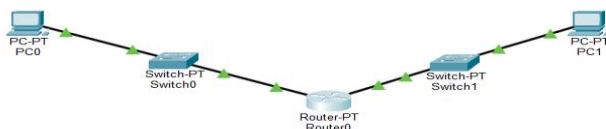


Fig 2.1 complete Packet Tracer topology

EXPLANATION:

The two LANs are connected through a router, with PC0 and PC1 placed in different IPv4 networks.

SCREENSHOT : IP Configuration

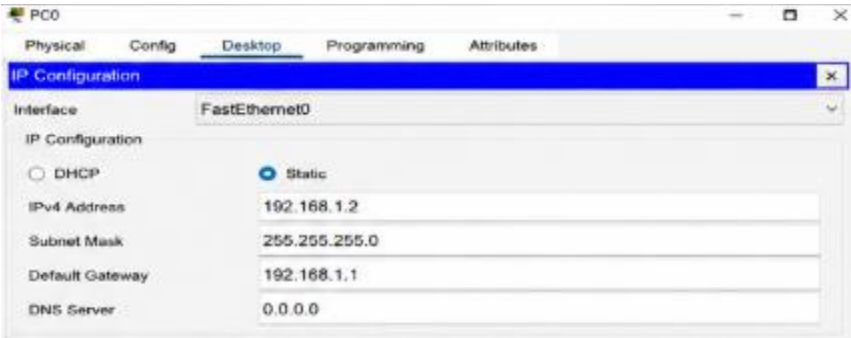


Fig 2.2 PC0 → Desktop → IP Configuration.

EXPLANATION:

PC0 is configured with an IPv4 address, subnet mask, and default gateway for the first LAN.

SCREENSHOT:

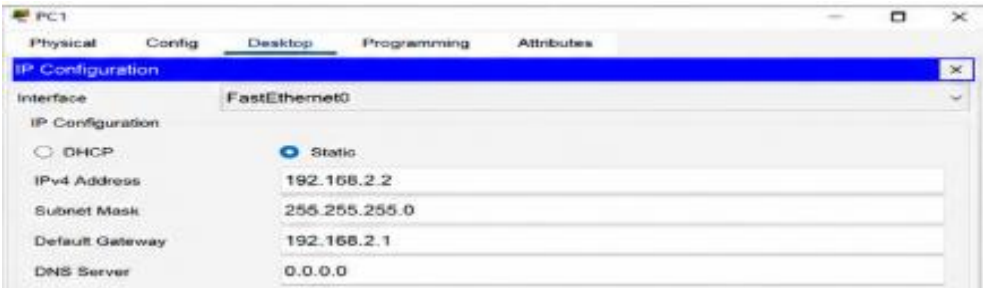


Fig 2.3 PC1 → Desktop → IP Configuration.

EXPLANATION:

PC1 is configured with an IPv4 address, subnet mask, and default gateway for the second LAN.

SCREENSHOT: PING TEST

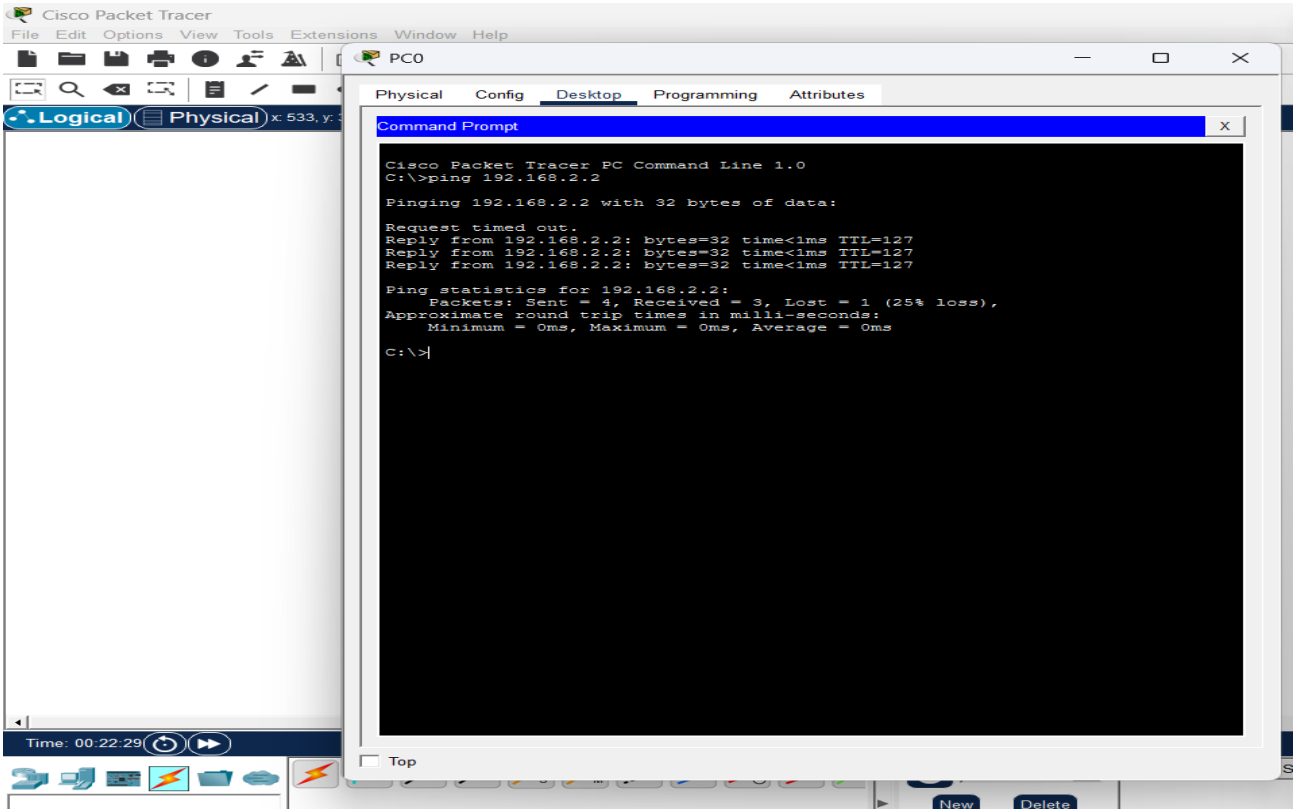


Fig 2.4 PC0 → Command Prompt showing

EXPLANATION:

The successful ping verifies end-to-end communication between the two different LANs through the router.

SCREENSHOT: Wireshark ipv4 header

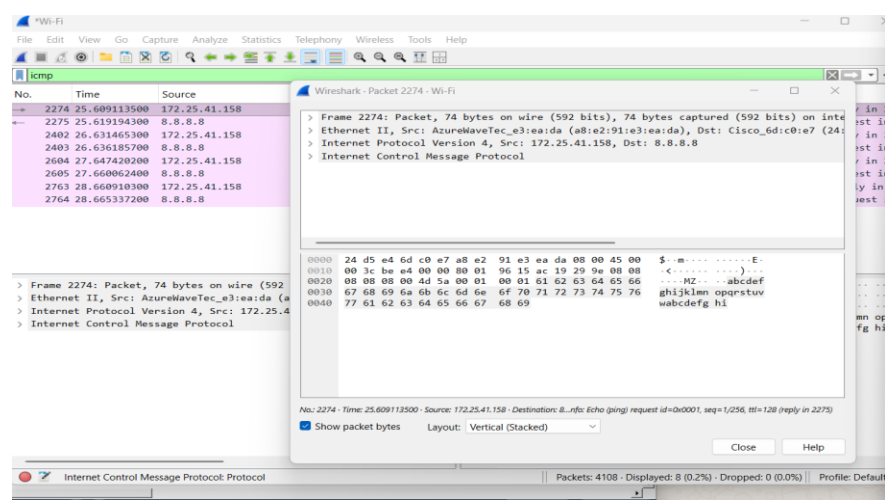


Fig: 2.5 Wireshark with filter

EXPLANANTION:

Wireshark captures the ICMP packet and allows analysis of the IPv4 header fields used for packet delivery between different networks.

RESULT: Thus, two IPv4 LANs were successfully connected using a router, and communication between PC0 and PC1 was verified using ICMP ping. The ICMP packet was captured using Wireshark, and the IPv4 header fields such as Source IP, Destination IP, TTL, Protocol, and Header Length were successfully analyzed.

3) AIM:

To construct a LAN using Cisco Packet Tracer, clear the ARP cache, perform a ping between hosts, and analyze ARP Request and ARP Reply packets using Packet Tracer and Wireshark.

SOFTWARE REQUIRED: cisco packet tracer, wireshark

PROCEDURE:

1. Open Cisco Packet Tracer.
2. Add 3 PCs and 1 switch.
3. Connect all PCs to the switch.
4. Assign IPs: PC0–192.168.1.10, PC1–192.168.1.11, PC2–192.168.1.12.
5. Open PC0 Command Prompt and check the ARP table using `arp -a`.
6. Clear the ARP cache using `arp -d` and verify it is empty.
7. Switch to Simulation Mode and enable ARP and ICMP.
8. From PC0, run `ping 192.168.1.11` and observe the packets.
9. Analyze the ARP Request and ARP Reply using PDU Information.
10. Open Wireshark, apply the arp filter, and analyze the captured ARP packets.

IMPLEMENTATION:

SCREENSHOT: Network Topology

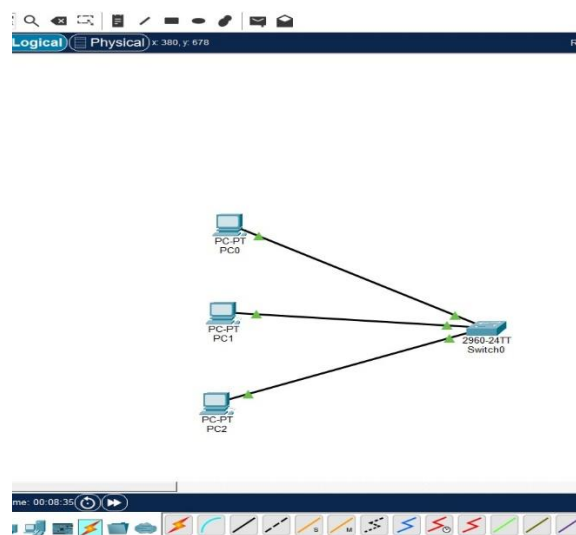


Fig 3.1: Network Topology

EXPLANATION:

The LAN is constructed using three hosts connected to a switch. Each host is configured with a unique IP address within the same subnet.

SCREENSHOT: ARP Cache Cleared

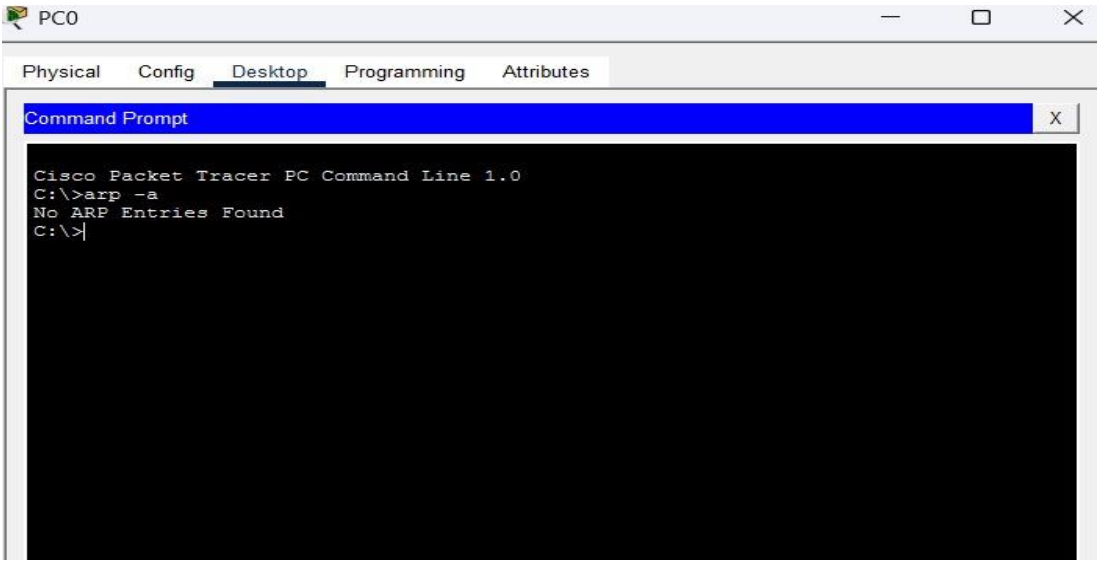


Fig 3.2 ARP cache cleared

EXPLANATION:

The ARP cache of PC0 is empty. Therefore, PC0 must perform a new ARP resolution to obtain the MAC address of PC1 before transmitting data.

SCREENSHOT : Ping and ARP Table

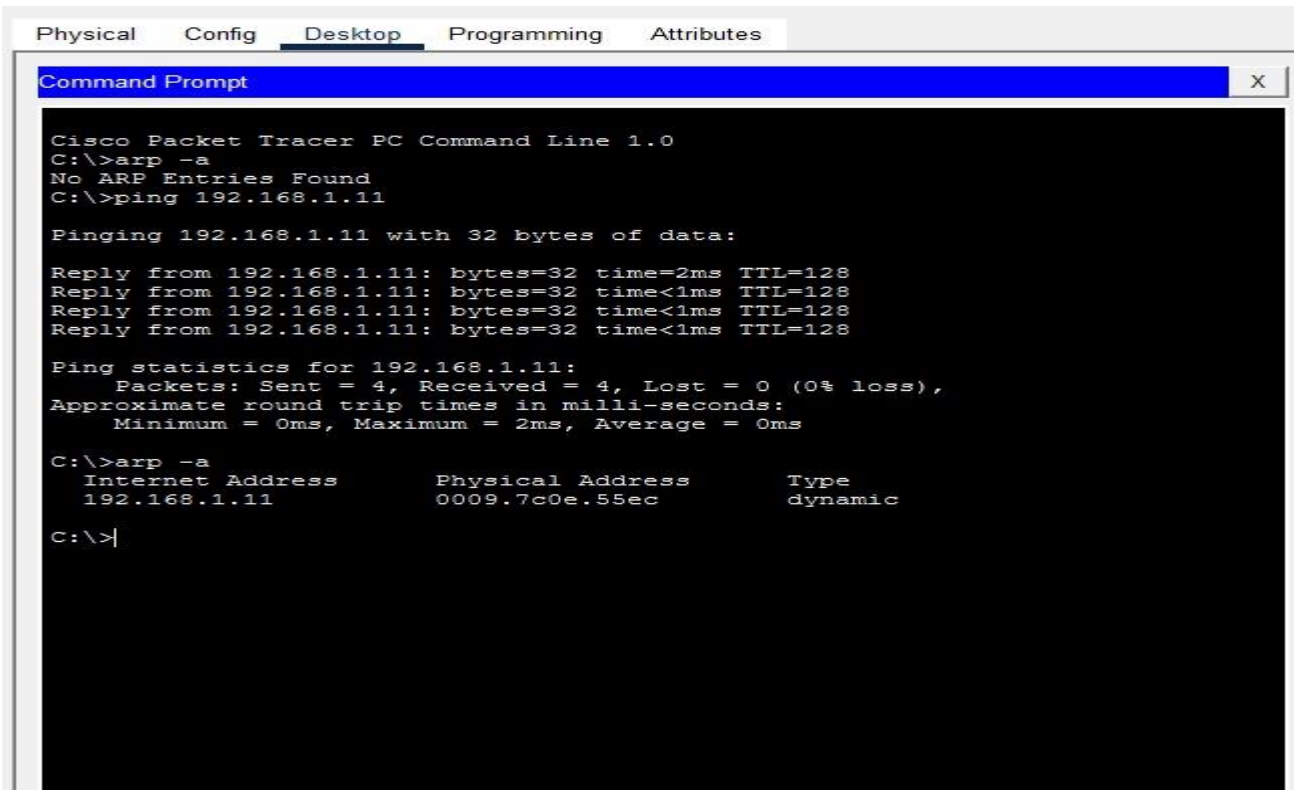


Fig3.3:Ping and ARP Table

EXPLANATION:

The ping between PC0 and PC1 is successful. After ARP resolution, PC0 stores PC1's IP-to-MAC mapping in its ARP cache

SCREENSHOT: ARP Request

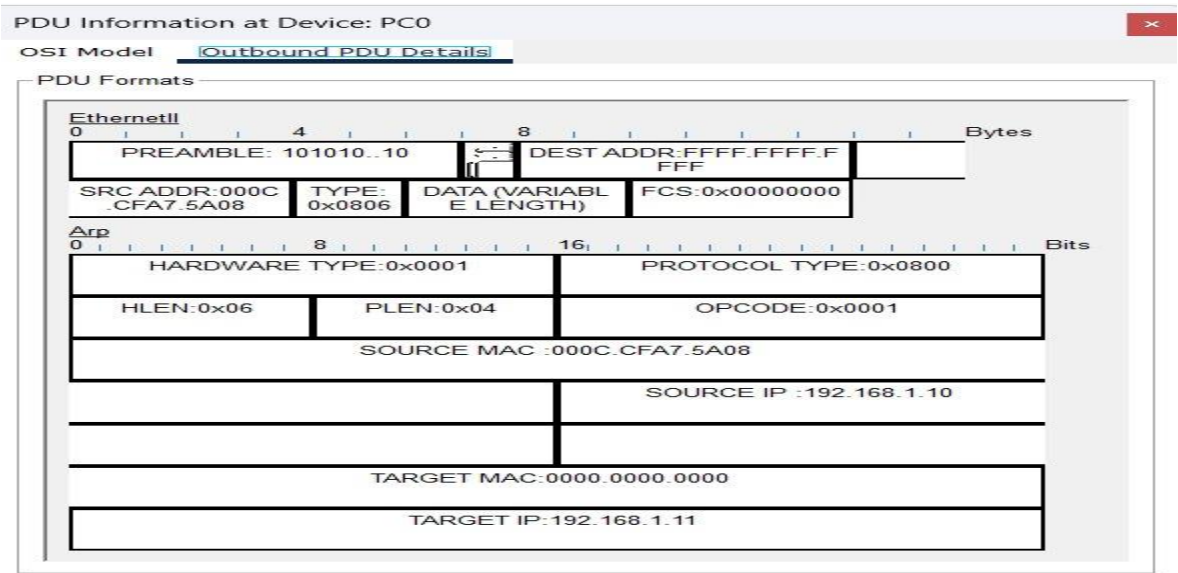


Fig 3.4 : ARP request

EXPLANATION:

PC0 broadcasts an ARP Request to find the MAC address associated with PC1's IP address, 192.168.1.11.

SCREENSHOT: ARP Reply / Wireshark Analysis

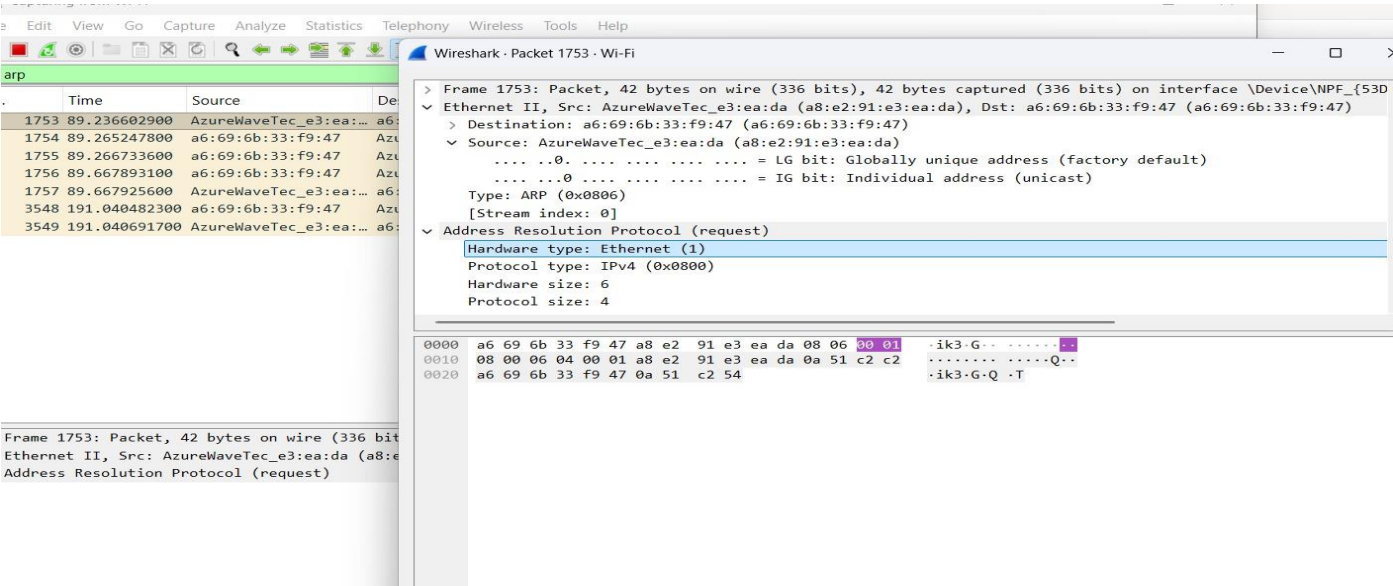


Fig 3.5:Wireshark Analysis

EXPLANATION:

PC1 sends an ARP Reply containing its MAC address to PC0. PC0 can then use this MAC address to deliver the ICMP data frames to PC1.

RESULT:

The LAN was successfully created, and the ARP Request and ARP Reply were captured and analyzed. ARP successfully resolved the IP address into the corresponding MAC address before data transmission.